Data types:-

It Represents the kind of data.

Single valued data-type

* int
* float
* bool
* complex
* str

multivalued datatypes (inbuilt data structure)

* list
* tuple
* set
* dict (dictionary)
* str

Example : $n_1 = 5$
type (n_1)

output : int

$n_1 = 5.6$
type (n_1)

output : tuple

name = "Rama kundella"
type (name)

output : str

value = 8.7
type (value)

output : float

value = 8.7 + 2j
type (value)

output : complex

value = True
type (value)

op : Bool

value = [1, "hi", 8.4]
type (value)

output : list

val = {1, "hi", 8}
type (value)

output : set

val = {1:1, 2:"hi", 3:8}

type(val)

output : dict

operators :

itc special symbol perform certine operation blw two operands.

Add = $n_1 + n_2$

operandas : n_1, n_2, Add

operators : = +

Types of operators :

1) Arithmatic operators : + - * / % ** //

a = 15	a = 10	a = 5
b = 10	b = 5	b = 8
a+b = 25	a-b	a * b
output = 25	= 5	= 45

2) Assignment operators : = += -= *= /= **= %= //=

n = 5	n+ = 4	n * = 3
n	n	n
op : 5	op : 9	op : 27

3) logical operators : AND OR NOT

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5) Membership operators: in not in

L1 = [3, 6, 7, 1, 8, 2] 5 not in L1 8 not in L1

5 in L1 false True false

6) Identity operator: is is not

a = 8 a = 8

b = 4 b = 10

a is b a is not b

false True

7) Bitwise operators: & | ~ >> <<

write a program to calculate simple interest and total Return value

```
pa = float(input("enter a principal Amount : "))
Rate = float(input("enter a Rate of Interest : "))
time = float(input("enter number of years :"))
si = (pa * time * rate)/100
print("the simple interest is :", si)
print("the total Amount Returned is :", si + pa)
```

conditional statements.

1) If condition

Syntax

```
If (condition):
print()
```

write a program to check the person is eligible for voting or not based on age

```
age = int(input("enter your Age:"))
If (age > 18):
    print("you are eligible for voting")
```

If else statements

Syntax :

```
if (condition):
    statements of if
else :
    statements of else
```

Ex:

```
age = int(input("enter your age:"))
if (age > 18):
    print("you are eligible for voting")
else :
    print("you are not eligible for voting")
```

write a program to check the given number is even or odd

```
num = int(input("enter a number :"))
if (num % 2 == 0):
    print("Its even number")
```